

JEE MAIN REVISION MASTER: COMPLEX NUMBERS 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Complex Numbers - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Complex Numbers - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Complex Numbers.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\sum_{n=1}^4 (-1)^{2n} \cdot i^{2n} = \dots$ " [Ans: C]
- [2025 Shift] Q#2 (MCQ): "If Z_1 and Z_2 are two non-zero complex numbers, then which of the following is not true?..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The real value of ' α ' for which $(1 - i \sin \alpha / 1 + 2i \sin \alpha)$ is purely real is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$\sum_{n=1}^4 (-1)^{2n} \cdot i^{2n} =$

- (A) 2
- (B) -i
- (C) 0 [Correct]
- (D) i

Correct Answer: C

Solution Analysis: $(-1)^{2n}$ is always equal to 1, because the power $2n$ is even. So, $(-1)^{2n} \cdot i^{2n} = 1 \cdot i^{2n} = i^{2n}$. Now use $i^2 = -1$. $i^{2n} = (i^2)^n = (-1)^n$. So the given sum becomes $\sum_{n=1}^4 (-1)^n = 1(-1)^1 + (-1)^2 + (-1)^3 + (-1)^4 = -1 + 1 - 1 + 1 = 0$. So the correct option is 0 Option C.